15

$N = 1000$

$M = 300$

$n = 4$

A

$$P(X=2) = \frac{\binom{300}{2}\binom{1000-300}{4-2}}{\binom{1000}{4}} = \frac{44850 \cdot \binom{700}{2}}{\binom{1000}{4}}$$

$$\boxed{P(X=2) \cong 0,265}$$

B Ahora hay independencia
Usamos Binomial

$n = 4$ $\qquad P = 300/1000$

$$P(X=k) = \binom{n}{k} p^k \cdot (1-p)^{n-k}$$

$$P(X=2) = \binom{4}{2} \cdot \left(\frac{3}{10}\right)^2 \cdot \left(\frac{7}{10}\right)^2 = 6 \cdot \frac{9}{100} \cdot \frac{49}{100} =$$

$$\boxed{P(X=2) \cong 0,2645}$$